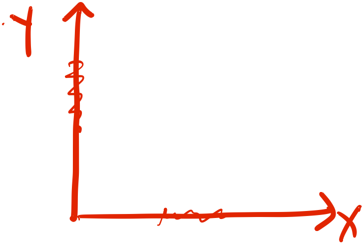
Elasticity

Elasticity

One of the challenges of the kind of demand analysis that we have discussed till now is that comparative analysis is challenging – different goods are measured in different units, and hence comparing outcomes across different goods is sometimes infeasible.



Price elasticity of demand

It is defined as the percentage change in quantity demanded in response to a one percent change in price.

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$$\text{Elasticity of demand } (e_d) = \frac{\text{\% change in quantity demanded}}{\text{\% change in price}} \quad \begin{matrix} \downarrow (-) \\ \uparrow (+) \end{matrix}$$

The elasticity of demand is negative in general - since the change in demand is opposite in direction to the change in price.

Price elasticity of demand

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$\downarrow 20\%$

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The elasticity of demand is negative in general - since the change in demand is opposite in direction to the change in price.

Interpretation: $e_d = -2$ implies that a $10\% \uparrow$ in prices leads to a $20\% \downarrow$ in demand. The negative value of elasticity implies that price and quantity move in opposite directions.

Price elasticity of demand

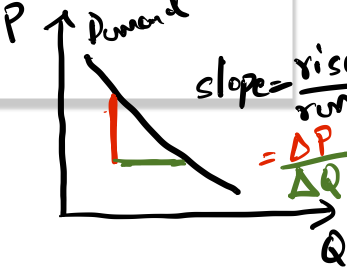
Dissecting the formula

$\Delta \rightarrow$ change.

$$e_d = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$$

$$e_d = \frac{\%\Delta Q^d}{\%\Delta P} = \frac{\frac{\Delta Q^d}{Q^d} \times \cancel{100}}{\frac{\Delta P}{P} \times \cancel{100}} = \frac{\Delta Q^d}{Q^d} \times \frac{P}{\Delta P}$$

$$= \underbrace{\frac{\Delta Q^d}{\Delta P}}_{1/\text{slope of dd}} \cdot \frac{P}{Q^d}$$



$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\Delta P}{\Delta Q}$$

Price elasticity of demand

Dissecting the formula

$$e_d = \frac{\text{\% change in quantity demanded}}{\text{\% change in price}}$$

$$= \frac{\%\Delta Q_x}{\%\Delta P_x} = \frac{\frac{\Delta Q_x}{Q_x} * 100}{\frac{\Delta P_x}{P_x} * 100} = \frac{\frac{\Delta Q_x}{Q_x} * \cancel{100}}{\frac{\Delta P_x}{P_x} * \cancel{100}}$$

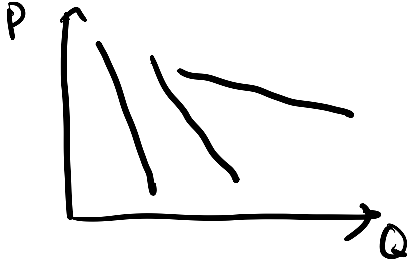
$$= \underbrace{\frac{\Delta Q_x}{\Delta P_x}}_{1/\text{Slope}} * \frac{P_x}{Q_x}$$

Price elasticity of demand

The case of linear demand curve.

$$e_d = \underbrace{\frac{\Delta Q_x}{\Delta P_x}}_{1/\text{Slope}} * \frac{P_x}{Q_x}$$

In case of a linear demand, the slope term is constant.



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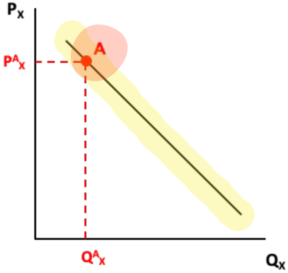

Let slope of this demand curve be s.

Then, e_d at A $= \underbrace{\frac{\Delta Q_x}{\Delta P_x}}_{= 1/s} * \underbrace{\frac{P^A_x}{Q^A_x}}_{\text{large number}}$

high / low

Price elasticity of demand

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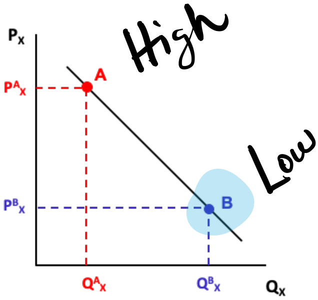

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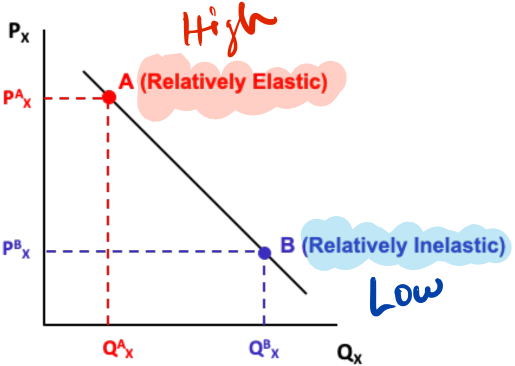
Then, e_d at B $= \underbrace{\boxed{\frac{\Delta Q_x}{\Delta P_x}}}_{=1/s} * \underbrace{\boxed{\frac{P^B_x}{Q^B_x}}}_{\text{small number}}$ low high

Price elasticity of demand

Linear demand curve

$$e_d = \underbrace{\frac{\Delta Q_x}{\Delta P_x}}_{1/\text{Slope}} * \frac{P_x}{Q_x}$$

In case of a linear demand, the slope term is constant.



Price elasticity of demand

Interpreting different values

-2, -3, -4 ...

- $e_d < -1$: Elastic demand.
- $e_d > -1$: Inelastic demand. → -0.9, -0.8, -0.7 ...
- $e_d = -1$: Unit elastic demand.

Price elasticity of demand – Example (try yourself!)

$$e_d = \underbrace{\frac{\Delta Q_x}{\Delta P_x}}_{1/\text{Slope}} * \frac{P_x}{Q_x}$$

If demand curve is given by $Q = 100 - 2P$, then calculate the price elasticity of demand at the following prices: (i) $P = 10$, (ii) $P = 20$, (iii) $P = 40$.

Price elasticity of demand – Example (try yourself!)

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If demand curve is given by $Q = 100 - 2P$, then calculate the price elasticity of demand at the following prices: (i) $P = 10$, (ii) $P = 20$, (iii) $P = 40$.

Slope of the demand curve = -1/2

When $P = 10$, then $Q = 80$. Therefore, $e_d = \text{-}2 * \frac{10}{80} = \text{-}0.25$

When $P = 20$, then $Q = 60$. Therefore, $e_d = \text{-}2 * \frac{20}{60} = \text{-}0.67$

When $P = 40$, then $Q = 20$. Therefore, $e_d = \text{-}2 * \frac{40}{20} = \text{-}4$

Note that demand is more *elastic* at higher prices, while it is relatively *inelastic* at lower prices.

Demand elasticity

What does elastic, inelastic and unit elastic mean?

$$e_d = \frac{\text{\% change in quantity demanded}}{\text{\% change in price}} = \frac{\%\Delta Q_x}{\%\Delta P_x}$$

- *Elastic*: If $e_d < -1 \implies |e_d| > 1 \implies |\%\Delta Q_x| > |\%\Delta P_x|$.

$-2, -3, -4 \ldots$

$$\left| \frac{\%\Delta Q_x}{\%\Delta P_x} \right| > 1$$

$$\underbrace{|\%\Delta Q_x|}_{\text{higher than } 10\%} > \overbrace{|\%\Delta P_x|}^{10\%}$$

Demand elasticity

What does elastic, inelastic and unit elastic mean?

$$e_d = \frac{\text{\% change in quantity demanded}}{\text{\% change in price}} = \frac{\%\Delta Q_x}{\%\Delta P_x}$$

Qty. effect is higher

- *Elastic*: If $e_d < -1 \implies |e_d| > 1 \implies \underbrace{|\%\Delta Q_x| > |\%\Delta P_x|}$.
- *Inelastic*: If $e_d > -1 \implies |e_d| < 1 \implies |\%\Delta Q_x| < |\%\Delta P_x|$.
- *Unit elastic*: If $e_d = -1 \implies |e_d| = 1 \implies |\%\Delta Q_x| = |\%\Delta P_x|$.

Price effect is higher

Determinants of price elasticity of demand

many → highly elastic

- *Number of close substitutes*:

few → relatively inelastic

Determinants of price elasticity of demand

- *Number of close substitutes*: If a good has many substitutes, then an increase in the price of the good could lead to large change in demand (as consumers move away from the good), thus leading to higher elasticity (relatively elastic).

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- *Time horizon*: $\xrightarrow[\text{run}]{\text{short}}$ inelastic.

  long → elastic

Determinants of price elasticity of demand

- *Number of close substitutes*: If a good has many substitutes, then an increase in the price of the good could lead to large change in demand (as consumers move away from the good), thus leading to higher elasticity (relatively elastic).
- *Time horizon*: The more time consumers have to respond to a price change, the larger the potential demand response, thus leading to higher elasticity (relatively elastic).

Applying Microeconomics to the real world.

Imagine you are a consultant and Old Navy (the company) asks for your advice on how to raise revenues from their products. How do you go about approaching this problem?
Hint: Think of Total Revenue.

$$\text{Revenues} = P \times Q.$$

Revenues ↑

① ↑ ↓ Net effect = ?

② ↓ ↑ Net effect = ?

If elastic, Qty. effect is higher ②

If inelastic, Price effect is higher ①

Applying Microeconomics to the real world.

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f

What happens to total expenditure when price increases?

Total Exp. = P * Q || When demand is elastic.

If P $\uparrow$, then we know that Q $\downarrow$. But, what happens to total expenditure?

$$\underbrace{\text{Total Expenditure}}_{?} = \underbrace{P}_{\uparrow} * \underbrace{Q}_{\downarrow}$$

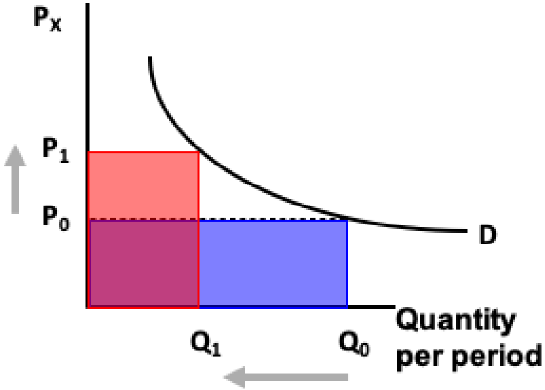
The net effect on total expenditure depends on which of the effects is stronger – the price effect or the quantity effect?

Since demand is assumed to be elastic, $e_d < -1$. Therefore, $e_d < -1$ $\implies |e_d| > 1) \implies |\%\Delta Q_x| > |\%\Delta P_x| \implies$ quantity effect is stronger.

$$\underbrace{\text{Total Expenditure}}_{\downarrow} = \underbrace{P}_{\uparrow} * \underbrace{Q}_{\downarrow}$$

What happens to total expenditure when price increases?

Total Exp. = P * Q || When demand is elastic. || Graphical approach



What happens to total expenditure when price of an inelastic good ↑ses?

revenue

Try yourself!

What happens to total expenditure when price of an inelastic good $\uparrow$ses?

Try yourself!

Now if $e_d > -1$ (or $|e_d| < 1$, inelastic demand). Then if price $\uparrow$, then Q $\downarrow$. But what happens total expenditure?

$$\underbrace{\text{Total Expenditure}}_{?} = \underbrace{P}_{\uparrow} * \underbrace{Q}_{\downarrow}$$

The net effect on total revenue depends on which of the effects - the price or the quantity effect, is stronger.

In this case, since $e_d > -1$, we know that $|\%\Delta Q_x| < |\%\Delta P_x|$, and hence the price effect is stronger. Therefore, the direction of the net effect is given by the price effect.

$$\underbrace{\text{Total Expenditure}}_{\uparrow} = \underbrace{P}_{\uparrow} * \underbrace{Q}_{\downarrow}$$

Income elasticity of demand

It is the percentage change in quantity demanded for a one percent change in income.

$$\text{Income elasticity of demand } (e_d^I) = \frac{\text{\% change in quantity demanded}}{\text{\% change in income}}$$

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The income elasticity of demand of a normal good is positive - since the change in demand is in the same direction as the change in income.

The income elasticity of demand for an inferior good is negative.

Cross-price elasticity of demand

It is the percentage change in quantity demanded of one good for a one percent change in price of another.

good x

$$\text{Cross-price elasticity } (e_d^{CP}) = \frac{\text{\% change in quantity demanded}}{\text{\% change in price of other good}}$$

good y

substitutes

$\uparrow P_y$ $\uparrow Q_x$

$(+)$

complements

$\uparrow P_y$ $\downarrow Q_x$

$(-)$

Cross-price elasticity of demand

It is the percentage change in quantity demanded of one good for a one percent change in price of another.

$$\text{Cross-price elasticity } (e_d^{CP}) = \frac{\text{\% change in quantity demanded}}{\text{\% change in price of other good}}$$

What would be the sign of e_d^{CP} for two goods that are substitutes?
Positive - since increase in price of one good leads to increased demand for the other good.

What would be the sign of e_d^{CP} for two goods that are complements?
Negative - since increase in price of one good leads to lower demand for the other good (as they are used together).

Consumer Surplus

Willingness to pay - Price

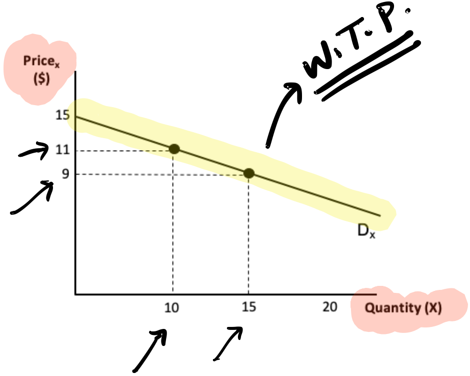
Consumer Surplus = W.T.P. minus Price.

$20 $12

gain $8.

Willingness to pay

Consumer's willingness to pay is measured using the demand curve.

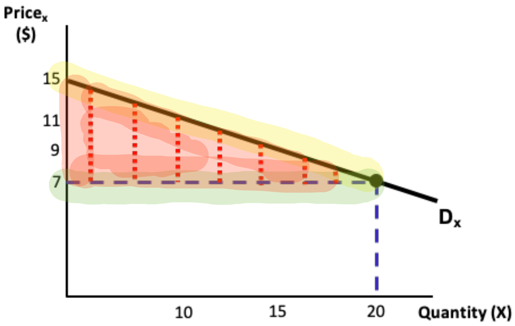


At 10 units of consumption the consumer is *willing to pay* \$11 for an extra unit of the good.

Similarly, at 15 units of consumption the willingness to pay is \$9.

Calculating consumer surplus (CS)

Consumer surplus = Willingness to pay (demand) minus price paid (equilibrium)



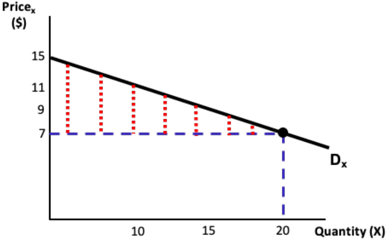
Therefore, the total consumer surplus is given by the sum of the CS derived by consuming units 1 through 20.

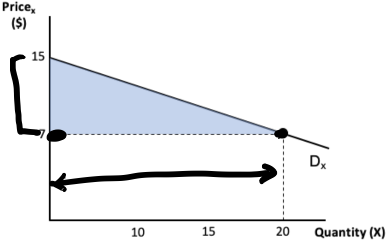
CS for a unit = red dotted line = willingness to pay (D_x) minus equilibrium price (price line)

Total CS = adding the CS by consuming each unit up to 20 units

Calculating consumer surplus (CS)

Consumer surplus = Willingness to pay (demand) minus price paid (equilibrium)





CS for a unit = red dotted line

Total CS = adding up all possible red dotted lines

Total CS = area of shaded triangle.

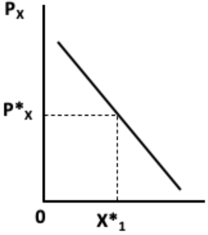
In this case, the area is given by $\frac{1}{2}\underbrace{(20)}_{\text{base}}\underbrace{(15-7)} = 80.$

Market demand curve

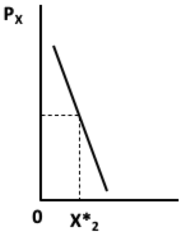
Consumer

Market demand curve

Till now, we have dealt with individual demand curves. What if we were interested in knowing the demand curve for the whole market?



(a) Individual 1



(b) Individual 2

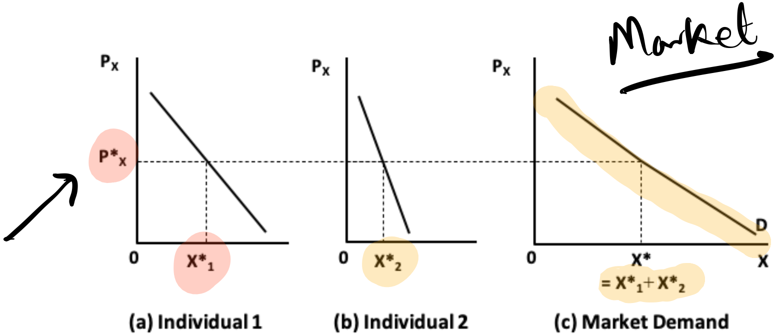
We can derive this by aggregating all the individual demand curves.

For the sake of simplicity, let there be two individuals in the market, and their demand curves are given in the figure to the left.

Deriving market demand curve

For the sake of simplicity, let there be two individuals in the market, and their demand curves are given below

Then the market demand curve is obtained by adding the quantity demanded of good x by each individual at a given price P_x.



How does market demand change when income increases?

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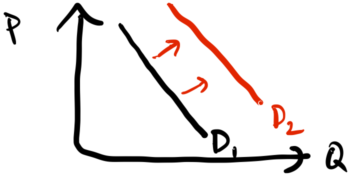
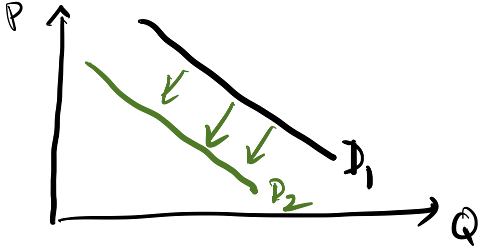


- In case the good is a normal good, then ↑ in income will lead to an ↑ (shift to the right) in demand for the good.
- In case the good is an inferior good, then ↑ in income will lead to a ↓ (shift to the left) in demand for the good.



How does market demand change when price increases?

Here we are referring to changes in prices of other goods.

TRY YOURSELF!

How does market demand change when price increases?

Here we are referring to changes in prices of other goods.

- In case the goods are complements, then an $\uparrow$ in price of one good will lead to a $\downarrow$ in demand (leftward shift) for the other.
- In case the goods are substitutes, then an $\uparrow$ in price of one good will lead to an $\uparrow$ in demand (rightward shift) for the other.